

BAAZ & UDAY JKBOSE

2026 CLASS 11TH

ROTATIONAL MOTION

LECTURE-1

PHYSICS

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Topics To Be Covered

- 1 Centre of Mass ✓
- 2 Why we use COM ✓
- 3 Centre of Mass of two-particle System ✓
- 4 Motion of COM ✓
- 5 Rotational Motion (x)

“BELIEVE IN YOURSELF”

**“EVEN WHEN NO ONE
ELSE DOES.”**

Syllabus

~~Imp~~

com $\rightarrow x | v | a \Rightarrow$ mid point formula

com $\rightarrow$ Rod | Sphere | Circle | Ring

Angular momentum L | Torque τ

~~Imp~~

Law of conservation of Angular momentum

Equilibrium

Momentum of Inertia.

Equation of Rotation Motion

Radius of Gyration

Comparison of linear Motion & Rotational Motion

System of Particles and Rotational Motion

Motion of System of Particles and Rigid Body

06 marks

Centre of mass of a two-particle system, momentum conservation and Centre of mass motion. Centre of mass of a rigid body, centre of mass of a uniform rod. Moment of a force, torque, angular momentum, law of conservation of angular momentum and its applications. Equilibrium of rigid bodies, rigid body rotation and equations of rotational motion, comparison of linear and rotational motions. Moment of inertia, radius of gyration, values of moments of inertia for simple geometrical objects (no derivation).

Recap From 1st Lecture

Work

- mode of transfer of energy

$$W = \vec{F} \cdot \vec{x}$$

$$W = Fx \cos \theta$$

Work

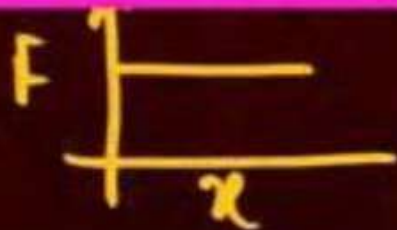
W = +ive

W = -ive

W = 0

Work

Work done by constant



$$W = \vec{F} \cdot \vec{x}$$

Work done by variable force



$$W = \int F \cdot dx$$

Energy

- Capacity to do work

$m \cdot E$

K.E

+

P.E

$$K.E = \frac{1}{2}mv^2$$

(Derivation)

✓
✓
✓
Detail
next
Lecture

Recap From 2nd Lecture

• Work-Energy Theorem

$$W = \Delta K \cdot E$$

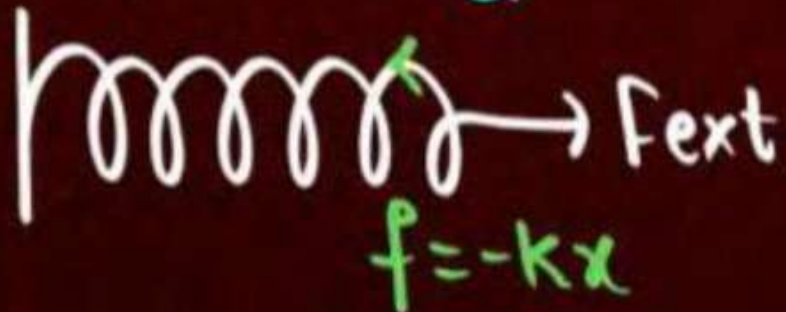
• P.E / U → Potential Energy

Zabardasti ✓

Relative position of particle

✓ Gravitation $P.E = mgh$

✓ Spring $P.E = \frac{1}{2} kx^2$



• WORK done by spring force

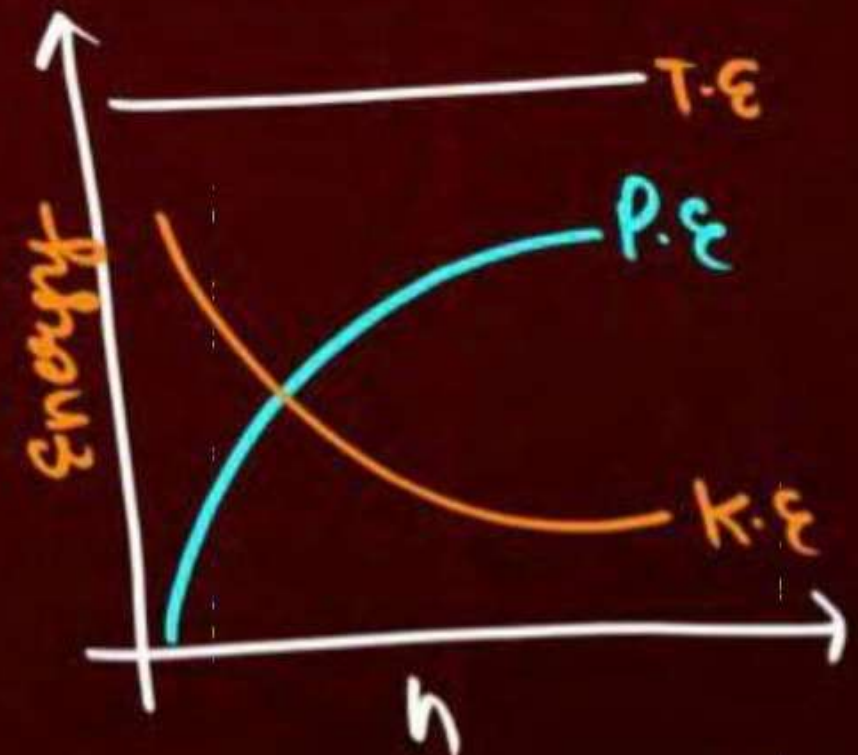
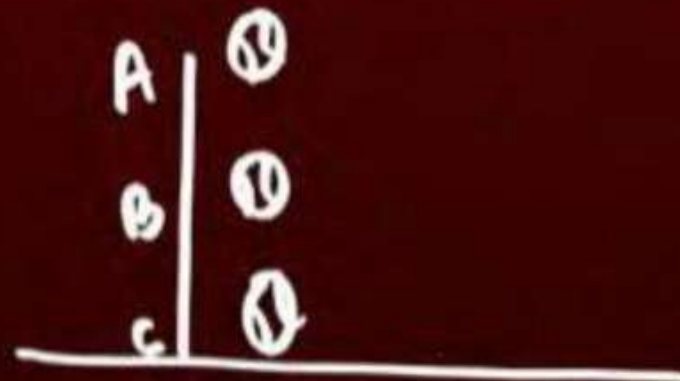
$$W = -\frac{1}{2} kx^2$$

Law of Conservation of Energy

$$T.E_i = T.E_f$$

$$m.E_i = m.E_f$$

100J 100J



Recap From 3rd Lecture

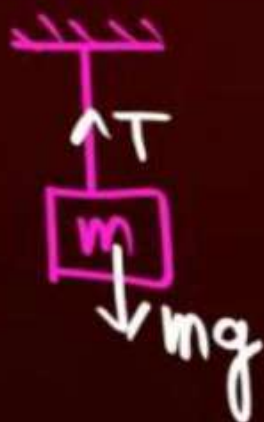
• Conservative Force

- Work $\downarrow$ ^{do not} depend on path.
- ✓ Gravitation Force ✓
- ✓ Spring / Electrostatic Force

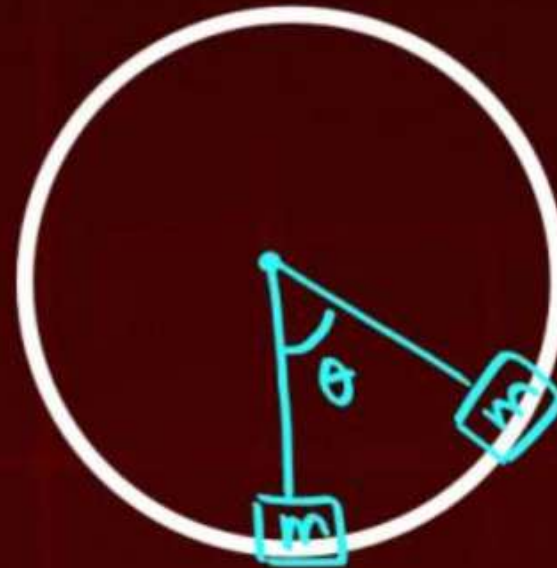
• Non-conservative force

- Work does depend on path
- ✓ Friction

Tension ✓



• Vertical Circular Motion



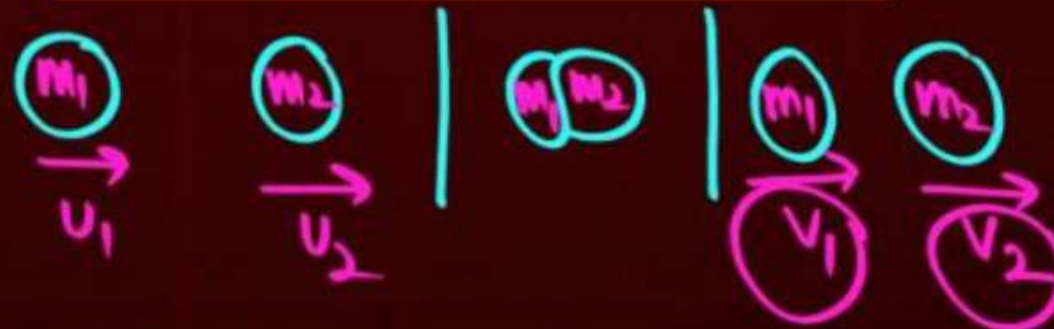
$$\checkmark \frac{mv^2}{R} = T - mg \cos \theta$$

$$\checkmark T = \frac{mv^2}{R} + mg \cos \theta$$

Collision

$P \rightarrow \text{cons.}$ $K, E \rightarrow \text{conserve}$

Perfectly elastic Collision

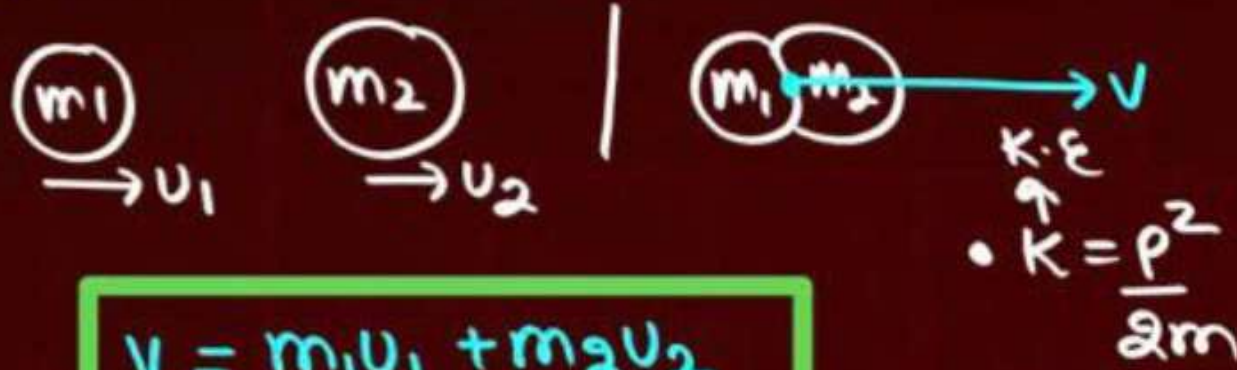


Perfectly inelastic collision

1 on 1

Recap From 4TH Lecture

Inelastic motion



$$v = \frac{m_1 u_1 + m_2 u_2}{m_1 + m_2}$$

$$p = \sqrt{2km}$$

Note

Power

$$P = \frac{W}{t} = Fv \cos \theta = \frac{K \cdot \epsilon}{t} = \frac{P \cdot \epsilon}{t}$$

✓ watt
✓ Hp

$$1 \text{ Hp} = 746 \text{ watt}$$

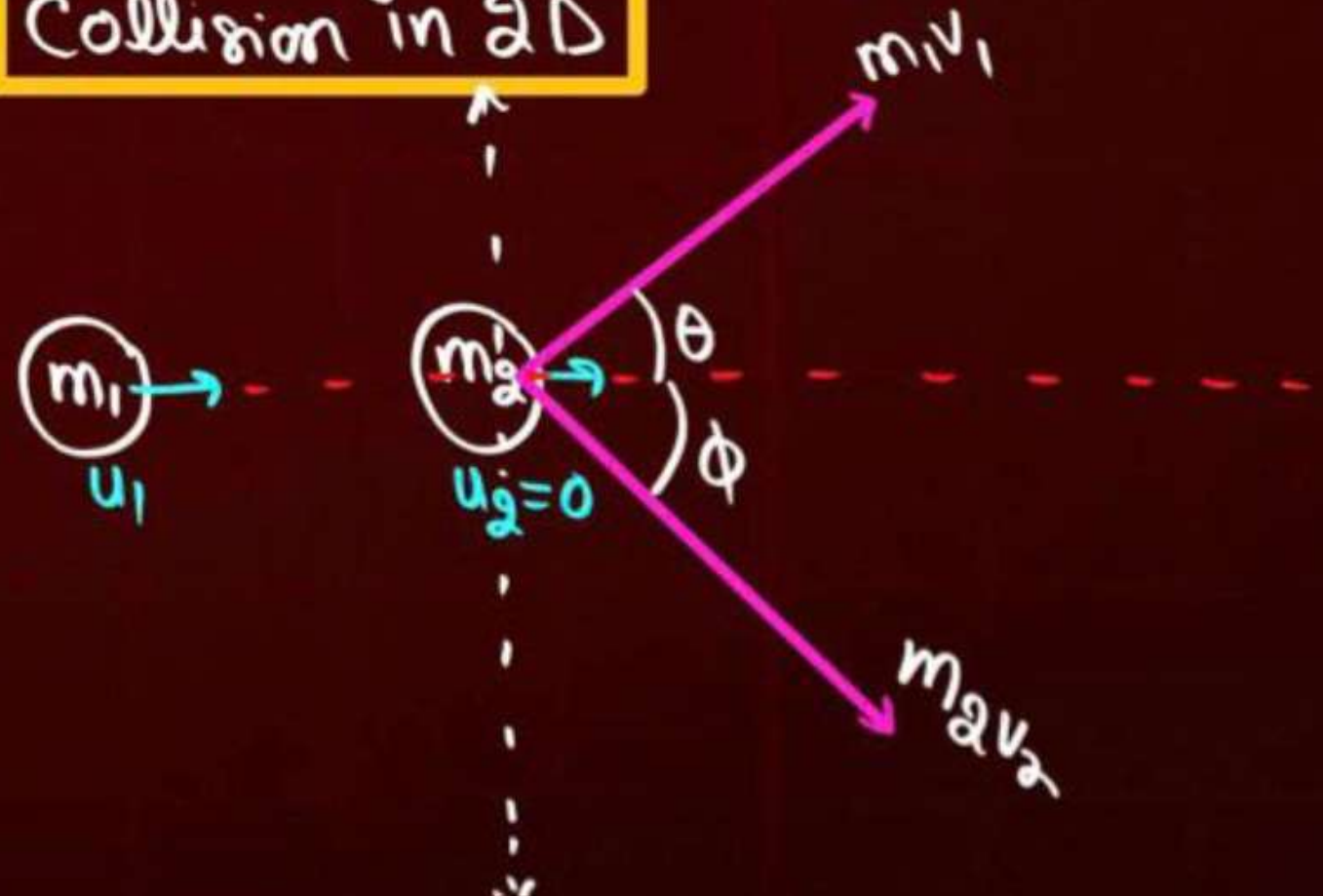
Coefficient of Restitution

$$e = \frac{v_2 - v_1}{u_1 - u_2}$$

$e = 0$ Perfectly inelastic collision

$e = 1$ Perfectly elastic collision

Collision in 2D



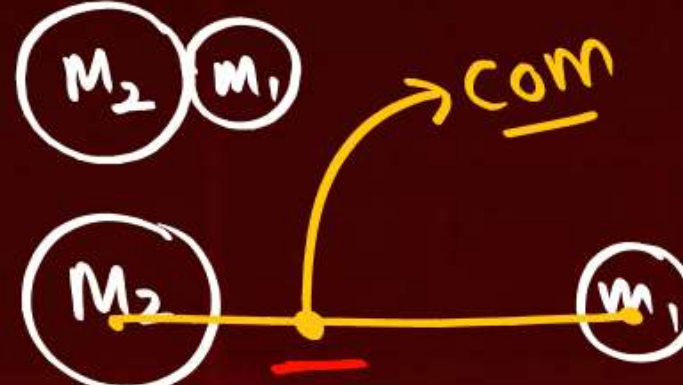
Centre of Mass

COM of mass is basically that point where the entire mass of system is assumed to be concentrated

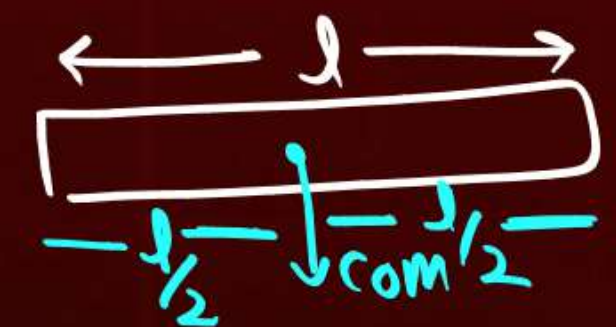


COM Important Point

① Imaginary point $\rightarrow$ where whole mass of Body is concentrated

②  M_2 m_1 com
 com is situated towards the Body with heavy mass

③ ~~com comes at a point where there is no mass.~~

④  l
 $\frac{l}{2}$ com $\frac{l}{2}$

⑥ Sphere

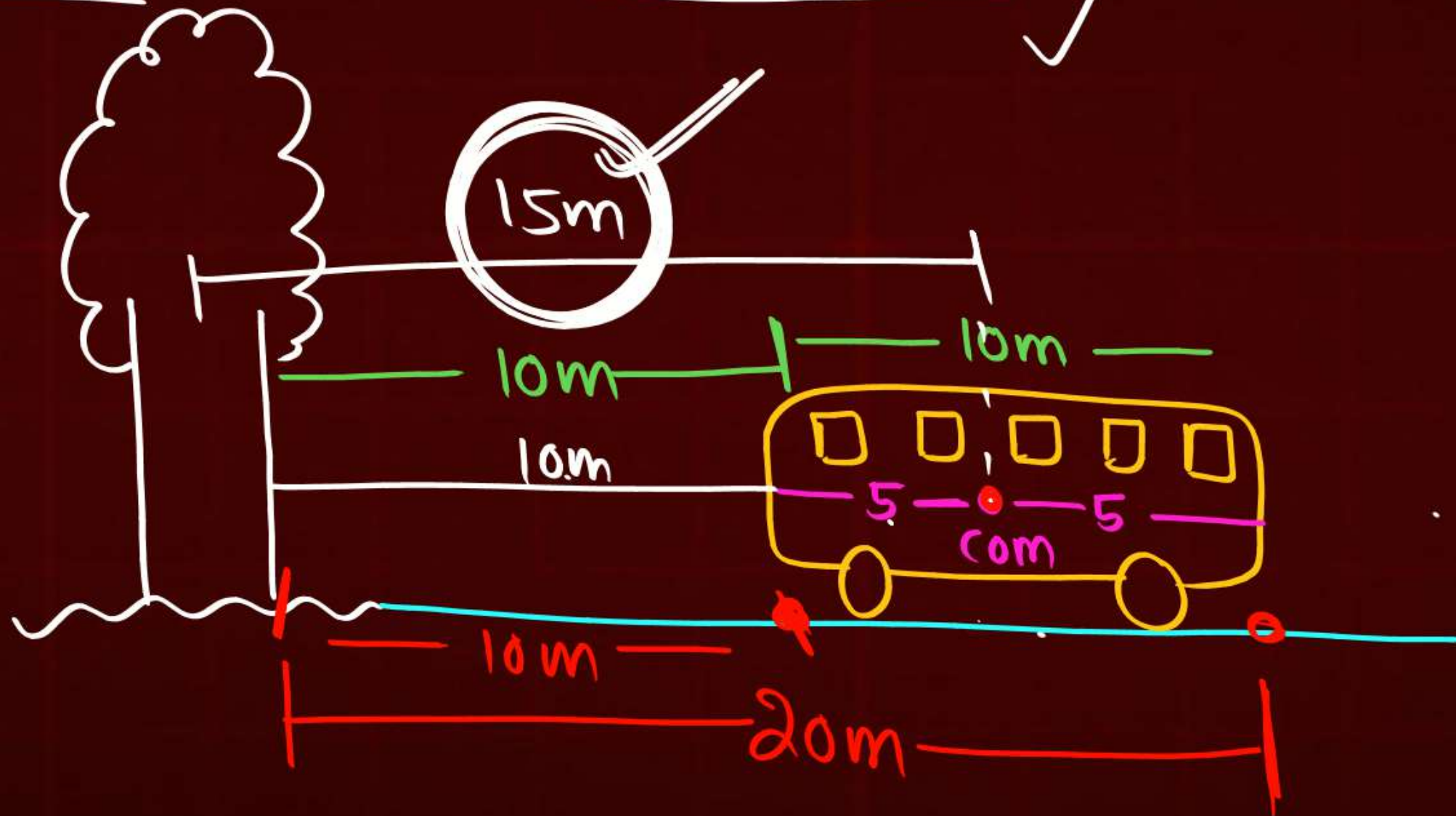


⑤ Ring (Bangle) $\rightarrow$ JKBOSE-2025)
 com

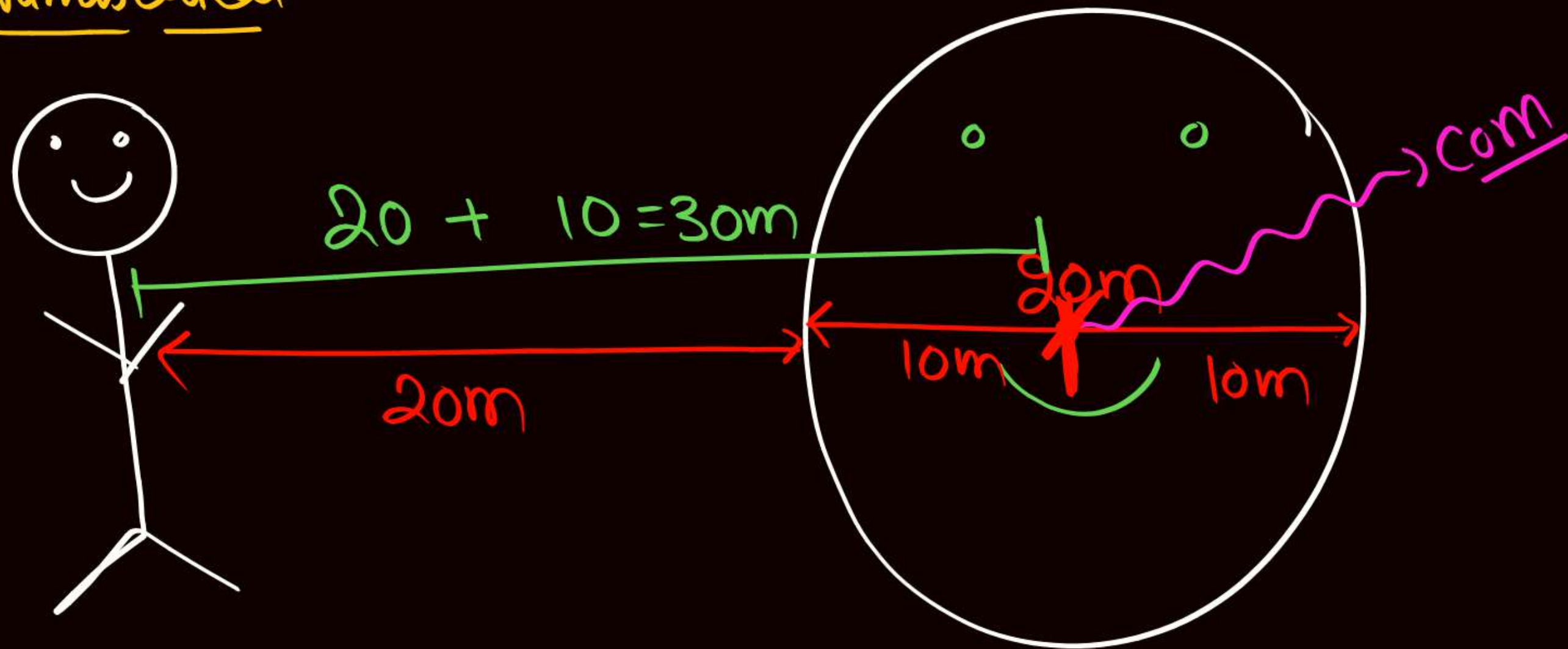
Why we use COM

(2016) JKBOSE

It makes it easy to solve mechanics problem where we have to describe the motion of oddly-shaped objects and complicated system.



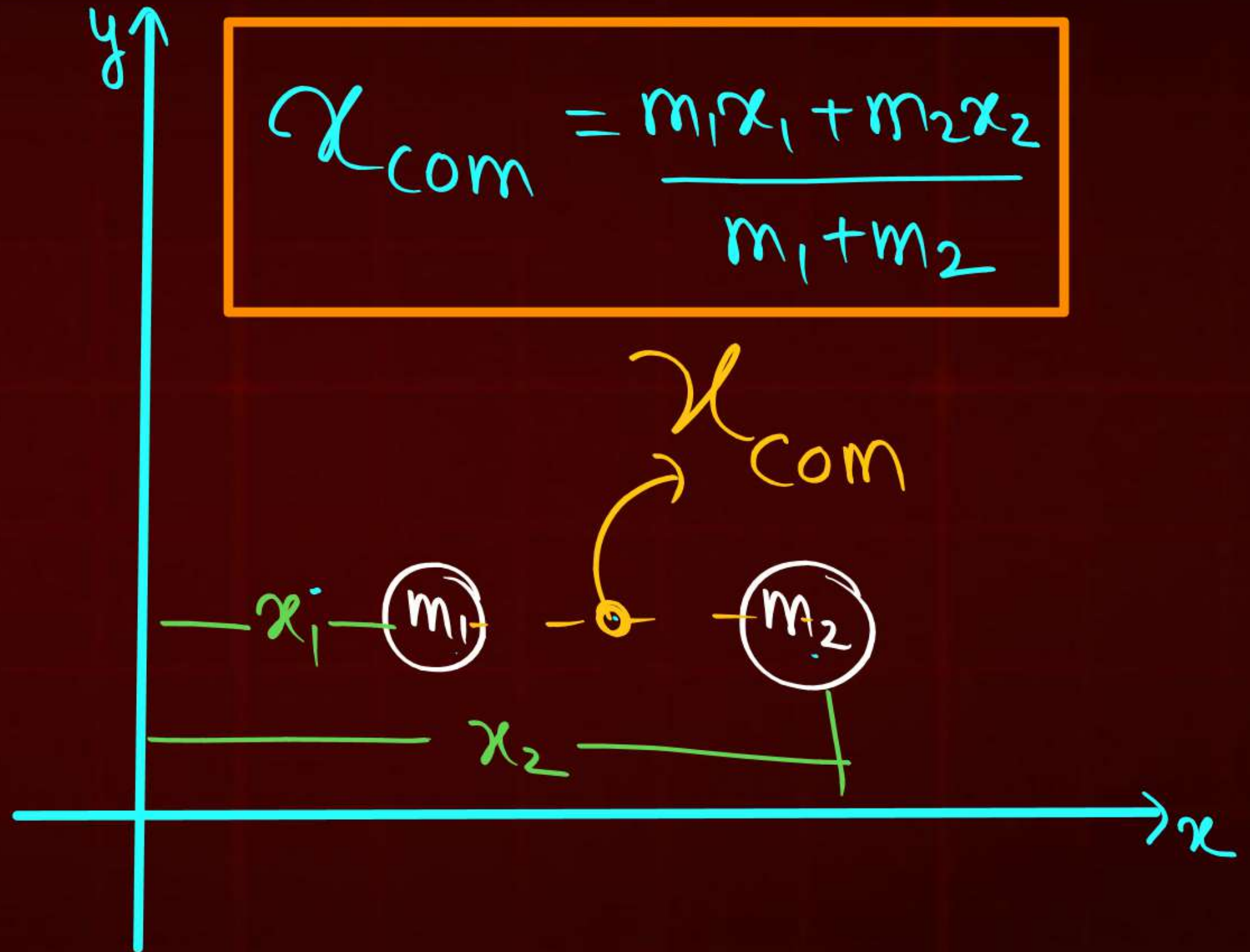
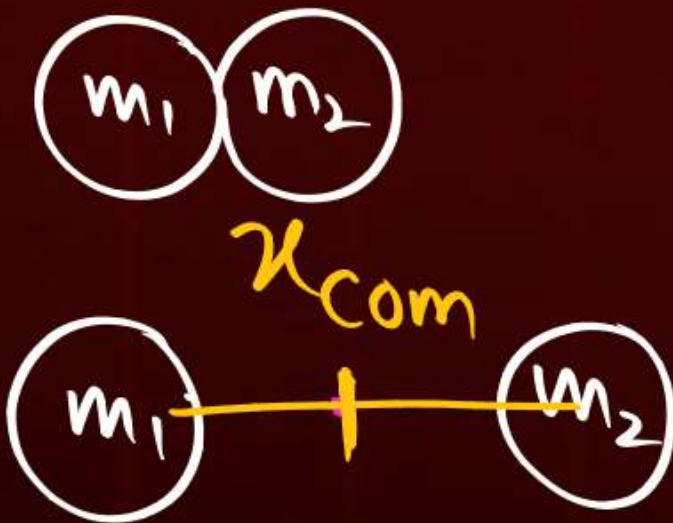
Q Numerical



Position of Centre of mass of two-particle system

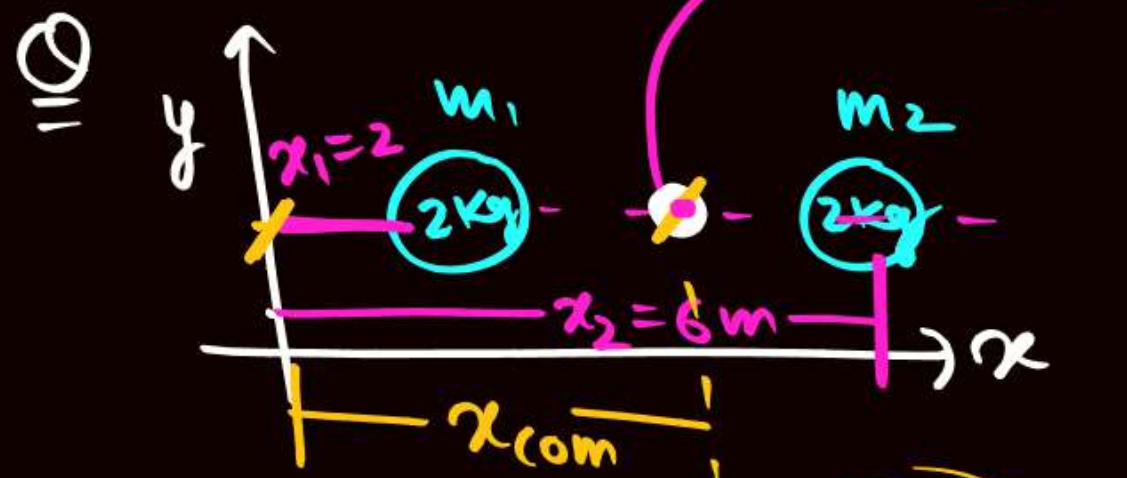
→ class 10th
mid point formula

let us consider
two body of
mass (m_1) and
 (m_2) situated
 (x_1) and x_2
from the origin



$$x_{com} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

Numerical



(a) 1 m → 4 m

(b) 2 m

(c) 3 m

(d) 4 m ✓

Sol

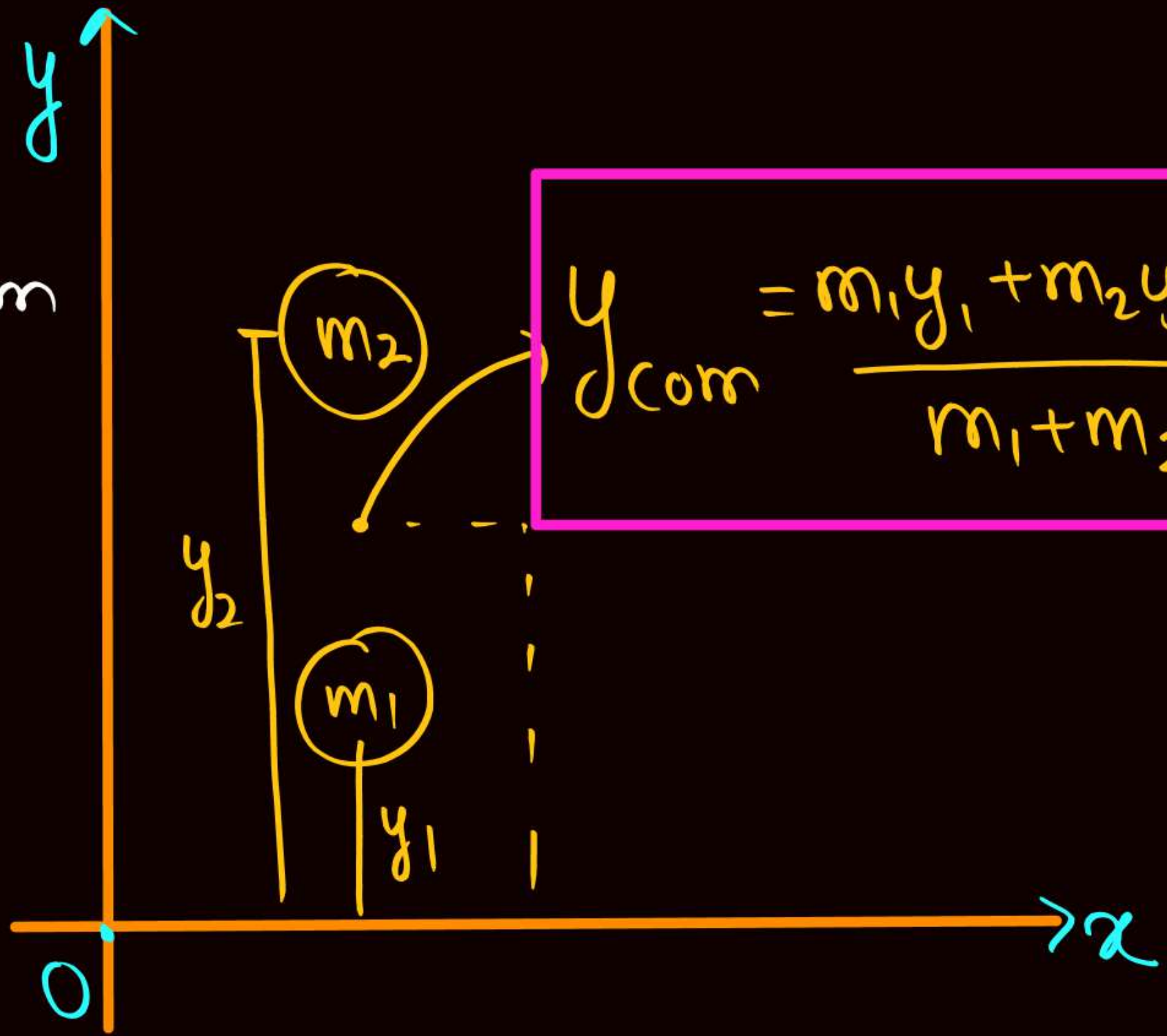
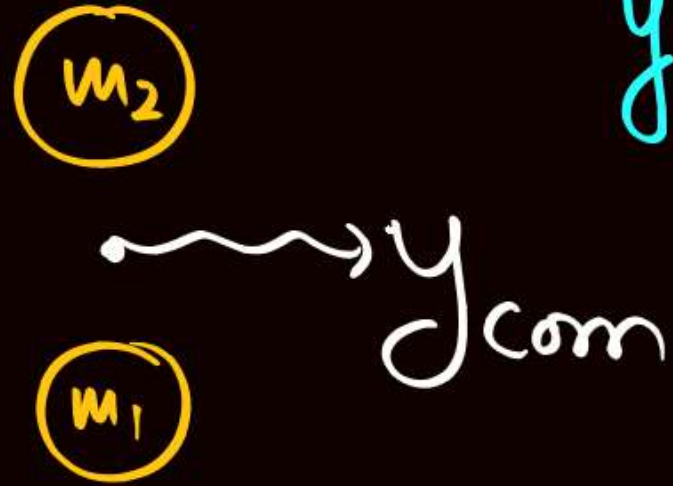
$$x_{\text{com}} = \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2}$$

$$= \frac{2 \times 2 + 2 \times 6}{2 + 2}$$

$$= \frac{4 + 12}{4}$$

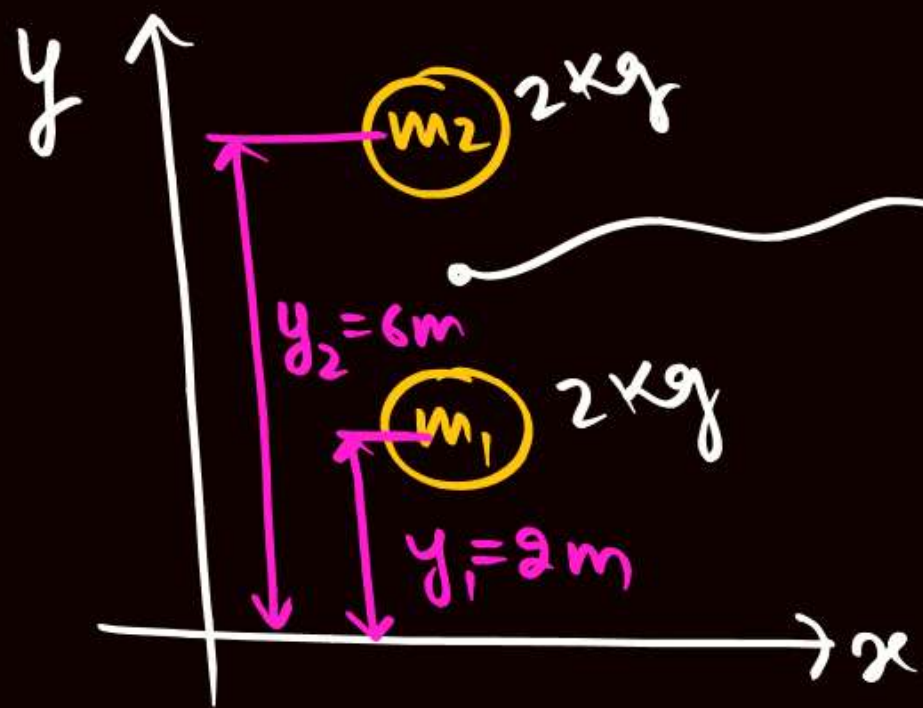
$$= \frac{16}{4} = 4\text{ m}$$

y_{com}



$$y_{com} = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2}$$

Numerical



$y_{\text{com}} = ?$

Sol

$$y_{\text{com}} = ?$$

$$y_{\text{com}} = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2}$$

$$= \frac{2 \times 2 + 2 \times 6}{2 + 2}$$

$$= \frac{4 + 12}{4}$$

$$y_{\text{com}} = \frac{16}{4} = 4\text{ m}$$

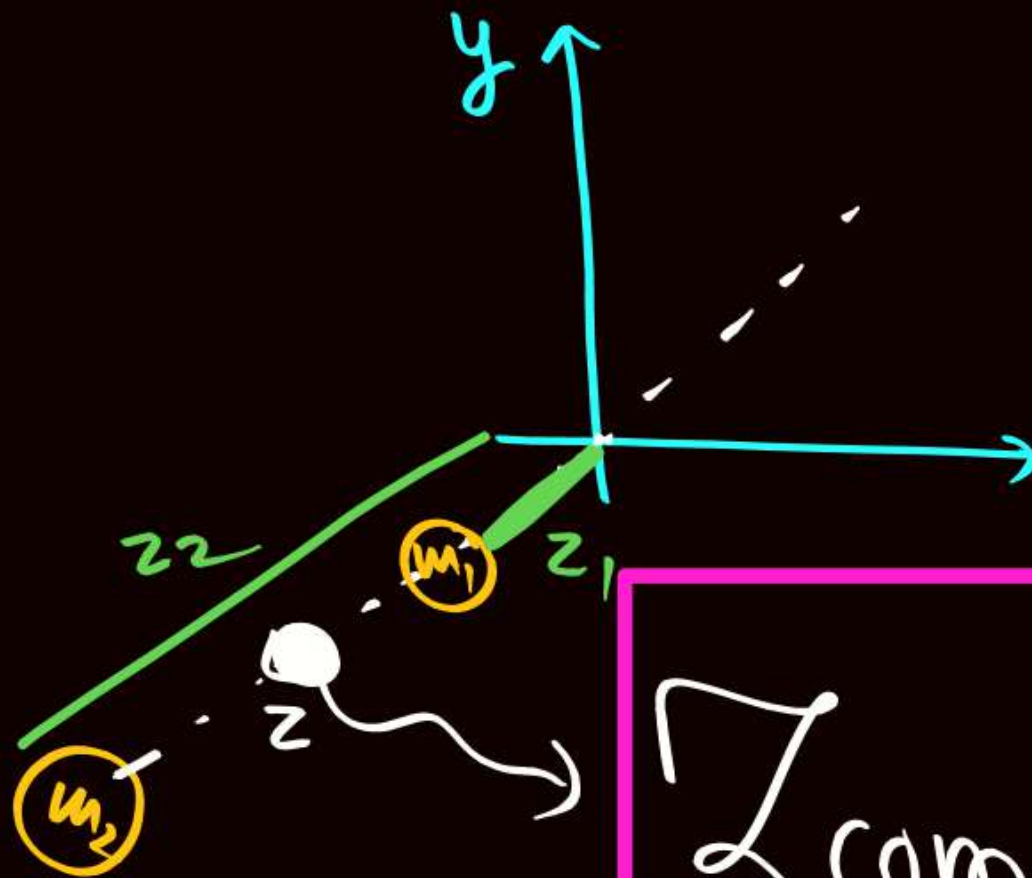
(a) 1

(b) 2

(c) 3

(d) 4

z_{com}

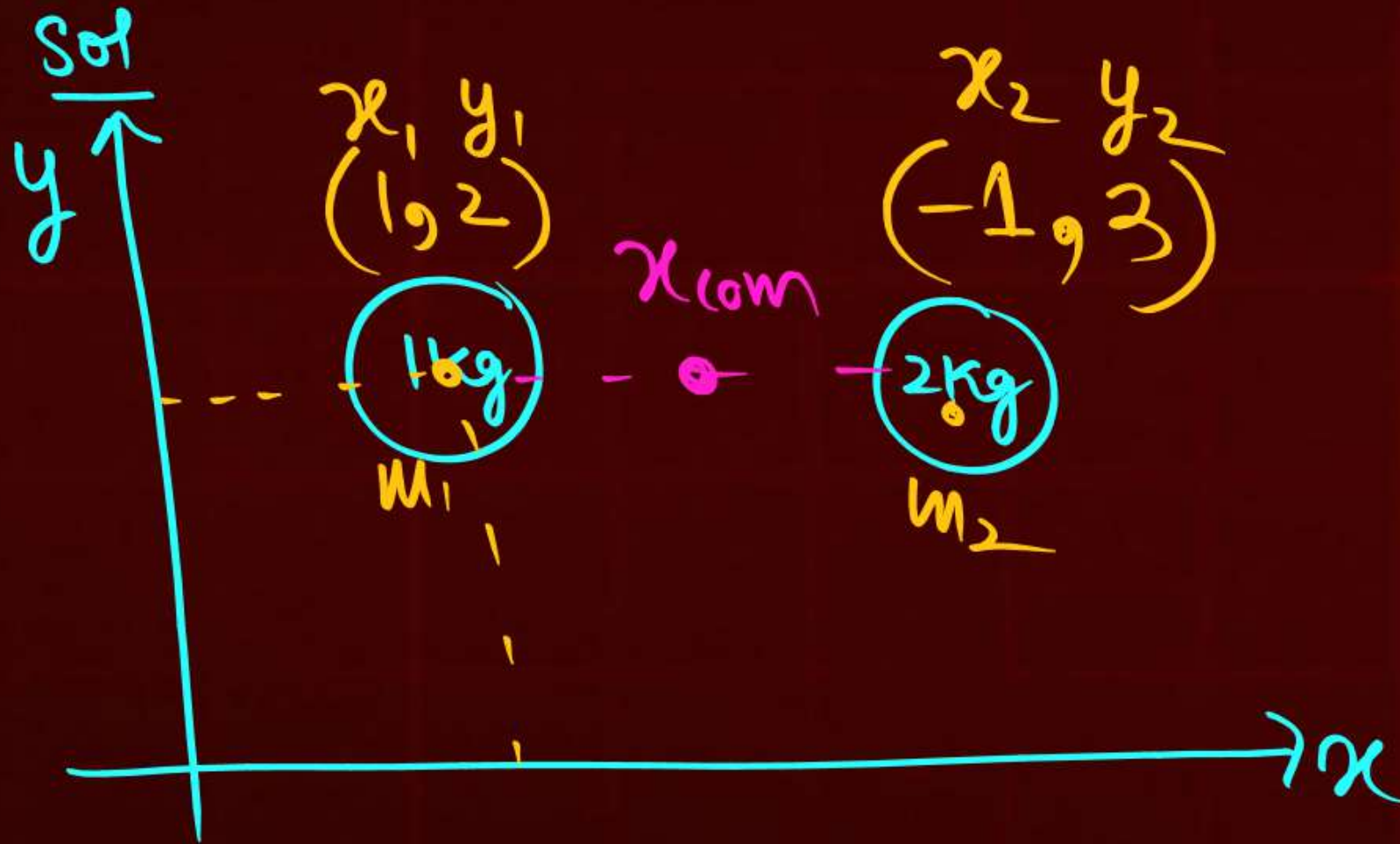


$$z_{com} = \frac{m_1 z_1 + m_2 z_2}{m_1 + m_2}$$

YE ASAAN HAI [SUMMER SERIES-A 2026]



Two bodies of masses 1 kg and 2 kg are located at $(1, 2)$ and $(-1, 3)$, respectively. Calculate the coordinates of the centre of mass. x_1, y_1 x_2, y_2
 x_{com} | y_{com}



$$\begin{aligned}x_{com} &= \frac{m_1 x_1 + m_2 x_2}{m_1 + m_2} \\&= \frac{1 \times 1 + 2 \times (-1)}{1 + 2} \\&= \frac{1 + -2}{3} = \frac{1 - 2}{3} \\x_{com} &= \frac{-1}{3} = \underline{\underline{-0.3}}\end{aligned}$$

$$y_{\text{com}} = \frac{m_1 y_1 + m_2 y_2}{m_1 + m_2}$$

$$= \frac{1 \times 2 + 2 \times 3}{1 + 2}$$

$$\underline{\underline{y_{\text{com}}}} = \frac{2 + 6}{3} = +\frac{8}{3}$$

$$x_{\text{com}} = -\frac{1}{3}$$

$$y_{\text{com}} = \frac{8}{3} \checkmark$$

YE ASAAN HAI [SUMMER SERIES-A 2026]

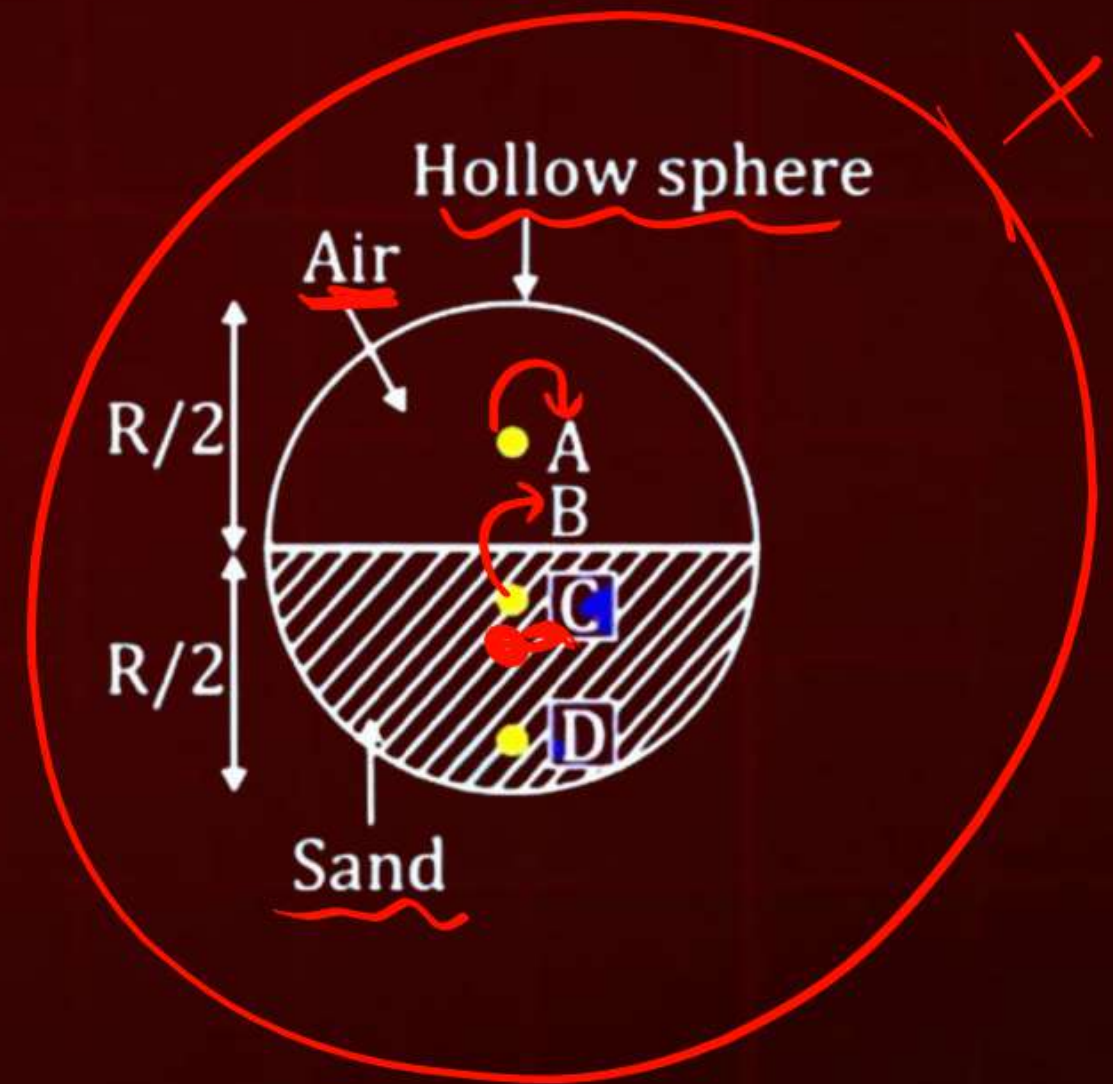
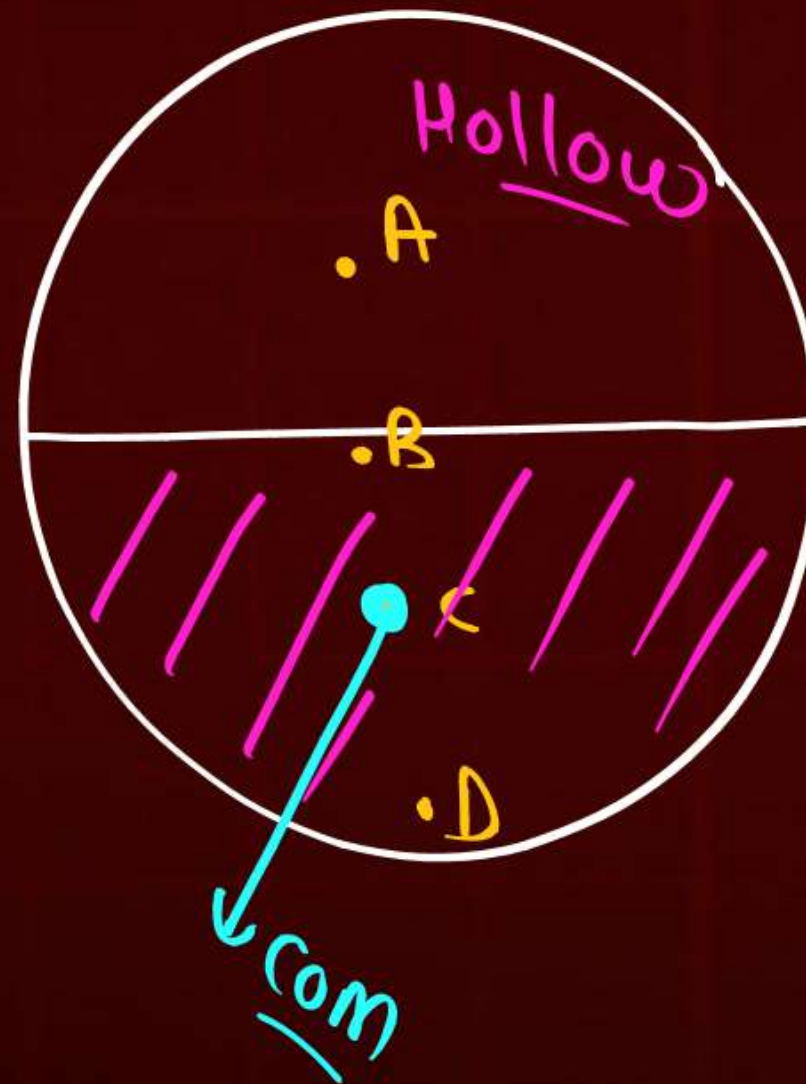
Which of the following points is the likely position of the centre of mass of the system shown in figure?

A A

B B

C C

D D



YE ASAAN HAI [SUMMER SERIES-A 2026]

✓ Is centre of mass a reality?

Answer:

No, the centre of mass of a system is a hypothetical point which acts as a single mass particle of the system for an external force.

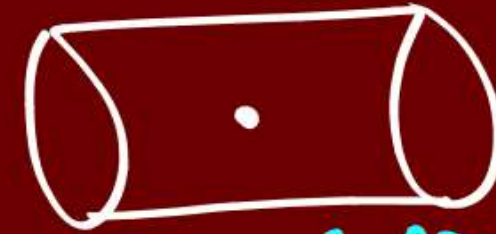
YE ASAAN HAI [SUMMER SERIES-A 2026]

Give the location of the centre of mass of a

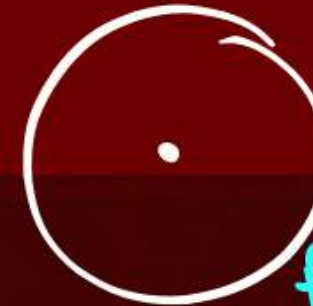
- (i) Sphere
- (ii) Cylinder
- (iii) Ring and
- (iv) Cube each of uniform mass density.



Sphere



Cylinder



Ring



Cube

Answer:

- (i) Centre of mass of a sphere lies at its geometrical centre.
- (ii) Centre of mass of a cylinder lies at its geometrical centre, i.e. at the mid-point of its axis of symmetry.
- (iii) Centre of mass of a ring lies at its geometrical centre.
- (iv) Centre of mass of a cube lies at its geometrical centre, i.e. at the point of intersection of its diagonals.

$$\vec{x}_{\text{com}} = \frac{m_1 \vec{x}_1 + m_2 \vec{x}_2}{m_1 + m_2}$$

$$\vec{y}_{\text{com}} = \frac{m_1 \vec{y}_1 + m_2 \vec{y}_2}{m_1 + m_2}$$

$$\vec{z}_{\text{com}} = \frac{m_1 \vec{z}_1 + m_2 \vec{z}_2}{m_1 + m_2}$$

General formula

$$\vec{r}_{\text{com}} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2}$$

YE ASAAN HAI [SUMMER SERIES-A 2026]

The position vectors of two points masses 10 kg and 5 kg are $(-3\hat{i} + 2\hat{j} + 4\hat{k})\text{m}$ and $(-3\hat{i} + 6\hat{j} + 5\hat{k})\text{m}$ respectively. Locate the position of centre of mass.

$$m_1 = 10\text{kg}$$

$$m_2 = 5\text{kg}$$

$$\vec{r}_1 = -3\hat{i} + 2\hat{j} + 4\hat{k}$$

$$\vec{r}_2 = -3\hat{i} + 6\hat{j} + 5\hat{k}$$

$$\vec{r}_{\text{com}} = \frac{m_1\vec{r}_1 + m_2\vec{r}_2}{m_1 + m_2}$$

SUMMARY



Com

Imp

x_{com}

y_{com}

z_{com}

$$\vec{r}_{\text{com}} = \frac{m_1 \vec{r}_1 + m_2 \vec{r}_2}{m_1 + m_2}$$

"BELIEVE IN YOURSELF"

**"EVEN WHEN NO ONE
ELSE DOES."**

Thank
You

